

## Unit 6 – Inference for Categorical Data: Proportions

### Notes 2 – Margin of Errors

#### Important Notes on Confidence Intervals

- “Margin of Error” does not mean that a mistake was made in data collection; it accounts for the \_\_\_\_\_ that exists when taking a sample from a population.
- Large samples are always better, however, if the sample size is large relative to the population, the proper inference we should use will be different than those taught in AP Statistics
- You must always name your inference procedure either by name (\_\_\_\_\_) or by the generic formula.

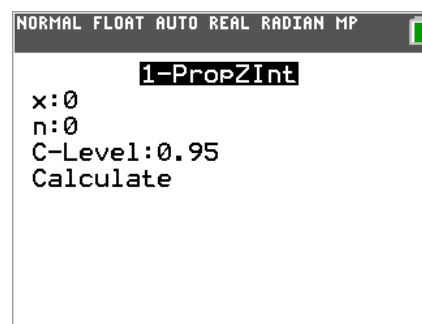
#### Confidence Intervals on the TI-84

STAT → Tests → A:1-Prop Z Int

x:

n:

C-Level:



#### Example: Using the Calculator

A poll was conducted at a large university to see if students preferred virtual classes vs in-person classes. The poll found that 175 of the 340 randomly selected university students said they preferred virtual classes.

a) What is the 99% confidence interval for the population proportion? (Assume all conditions are met)

Interval:

point estimate:

margin of error:

b) Interpret this interval in context.

c) Interpret the confidence level in context.

## Margin of Error

point estimate  $\pm$  margin of error

\_\_\_\_\_  $\pm$  \_\_\_\_\_

|                              |                              |
|------------------------------|------------------------------|
| Increasing Confidence Level: | Decreasing Confidence Level: |
|                              |                              |
| Increasing Sample Size:      | Decreasing Sample Size:      |
|                              |                              |

Keep In Mind:

- The margin of error in a confidence covers only accounts for \_\_\_\_\_
- The margin of error does not account for bias in the sampling methods

## Sample Size and Margin of Error

Often time, researchers need to both limit the margin of error to a fixed amount and still maintain a certain confidence level. In such cases, the only thing that can vary is the sample size, since the value for  $\hat{p}$  is something that is out of our control.

If our margin of error and our confidence level (therefore our critical level), we can solve for  $n$  to determine the sample size needed to obtain a certain margin of error size.

HOWEVER, we don't know  $\hat{p}$  until we actually conduct the study

We can then do one of two things:

- 1) Use an estimated  $\hat{p}$  based on previous studies (this would be the wording used in the problem)
- 2) Use a "conservative"  $\hat{p} = 0.5$ . Using a  $\hat{p}$  of 0.5 gives the largest possible margin of error for any given  $z$  or  $n$ . We will use this option when a value for  $\hat{p}$  is not given in the problem.

**Example: Finding Sample Size**

Before conducting nationwide polling prior to a national election, most good companies will decide ahead of time how many people to include in the poll based on a particular margin of error and confidence interval. Let's say that Gallup wants to conduct a poll to see what proportion of the voters would vote for Trump again if Donald Trump were running for president against Joe Biden. Gallup wants to be 95% confident with a margin of error no more than 3 percentage points. How many people must they randomly sample in order to achieve this?